Ref: HRSU/TDS/CNT/001 **Date: 09/10/2026**

TECHNICAL DATA SHEET

Calcium Nitrate Tetrahydrate — Industrial Grade

Product Information

Product Name	Calcium Nitrate Tetrahydrate	**Package**	50 kg PP woven bag with inner plastic liner
Assay	99 – 103% as $Ca(NO_3)_2 \cdot 4H_2O$	**Minimum Order**	10 MT
CAS No.	13477-34-4	**HS Code**	28342990
EC No.	233-332-1	**Molecular Formula**	$Ca(NO_3)_2 \cdot 4H_2O$
Grade	Industrial Grade	**Molecular Weight**	236.15 g/mol
Appearance	Colourless to white crystals	**Density (typical)**	Approx. 1.86 g/cm^3
Application	Wastewater treatment (H_2S and odour control), agriculture, chemical, mining	**pH (5% solution, 25 °C)**	5.0 – 7.0
Documents	TDS, SDS and COA	**Sample**	Available

Composition (calculated for pure tetrahydrate)

Component	Content, % w/w (calculated)
Calcium nitrate, $Ca(NO_3)_2$	Approx. 69.5
Water of crystallisation	Approx. 30.5
Nitrate, NO_3	Approx. 52.5
Nitrogen, N	Approx. 11.9
Calcium, Ca	Approx. 17.0
Calcium oxide equivalent, CaO	Approx. 23.7

The 30% water is chemically bound as water of crystallisation, not free moisture. These values are calculated from the formula and are not test results.

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Specification

Item	Specification	Method
Appearance	Colourless to white crystals	Visual
Assay (as $Ca(NO_3)_2 \cdot 4H_2O$)	99 – 103%	Titrimetric
Calcium (Ca), soluble	Min 16.4%	—
Nitrate nitrogen (N)	Min 11.7%	—
Melting point	42 – 44 °C	Capillary tube
pH (5% solution in water, 25 °C)	5.0 – 7.0	pH meter
Insoluble matter in water	Max 0.005%	Gravimetric
Nitrite (NO_2)	Max 0.001%	Colorimetric
Chloride (Cl)	Max 0.005%	Turbidimetric
Sulphate (SO_4)	Max 0.005%	Turbidimetric
Barium (Ba)	Max 0.01%	ICP-OES
Iron (Fe)	Max 0.0005%	ICP-OES
Lead (Pb)	Max 0.001%	ICP-OES
Magnesium (Mg)	Max 0.05%	ICP-OES
Potassium (K)	Max 0.005%	Flame photometry
Sodium (Na)	Max 0.02%	Flame photometry
Strontium (Sr)	Max 0.05%	ICP-OES
Solubility in water (typical)	121 g in 100 g of water	Visual

Storage, handling and safety

- The product is hygroscopic. Keep the bag tightly closed and store in a cool, dry, ventilated area, away from heat and sunlight.
- Keep away from organic materials, combustibles and reducing agents. Minimise dust and avoid contact with skin and eyes.
- Shelf life: 60 months from the date of manufacture when stored sealed and dry.
- Oxidising solid: UN 1454, Class 5.1, Packing Group III. See the Safety Data Sheet for full information.

Values are typical and given for guidance; batch-specific results are in the Certificate of Analysis. The dose for any application depends on site conditions and is determined by the user.

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